

FPT UNIVERSITY
ARTIFICIAL INTELLIGENCE DEPARTMENT



TRƯỜNG ĐẠI HỌC FPT

GRADUATION THESIS

**A gSASRec-Based Sequential Recommender System
for Next Learning Resource Prediction
on the MARS E-Learning Dataset**

Supervisor:	Dr. Nguyễn An Khương
Class:	AI18C
Students:	Nguyễn Hoàng Việt – QE180058 Nguyễn Tấn Thắng – QE180019 Lê Quốc Chính – QE170250

Quy Nhon, April 2026

Statement of Originality

We hereby declare that the graduation thesis entitled:

“A gSASRec-Based Sequential Recommender System for Next Learning Resource Prediction on the MARS E-Learning Dataset”

is the result of our original research conducted under the supervision of Dr. Nguyễn An Khương.

This work is developed based on prior studies, including the proposed gSASRec model by Petrov and Macdonald, together with other referenced publications cited appropriately throughout this thesis.

Except for materials explicitly acknowledged through citations and references, all analyses, implementations, experiments, and findings presented in this report are the original contributions of our research team.

Quy Nhon, April 2026

The Authors

Abstract

Online learning platforms provide learners with large catalogues of videos, courses, and instructional documents. In this setting, the main challenge is not only whether a learner may like a resource, but also which resource should be recommended next according to the learner’s recent learning sequence. This thesis studies next learning resource prediction on the MARS (Mandarine Academy Recommender System) e-learning dataset.

The current exploratory data analysis shows that the active interaction set contains 3,007 users, 958 learning resources, and 24,622 interactions, with 99.1453% sparsity. User sequences are short and unevenly distributed: the average user has 8.19 interactions, while only 1,676 users have at least three interactions and are therefore eligible for leave-one-out train/validation/test evaluation. The item metadata is also heterogeneous: software, theme, duration, and type fields have complete coverage, while difficulty is available for 40.70% of items and descriptions for 86.46%.

Objectives

The main objectives of this research are:

- To formalize the MARS recommendation task as a top- K next-item prediction problem.
- To build a reproducible preprocessing and evaluation pipeline for sparse e-learning interaction data.
- To compare simple and neural baselines against a gSASRec direction using Generalised BCE loss and multiple negative sampling.
- To analyze whether metadata-aware initialization, confidence weighting, and reranking can improve practical recommendation quality.

Methodology

Implicit and explicit interactions are merged into a unified active-interaction table. Explicit events with watch percentage greater than or equal to 50 are treated as active, implicit events are treated as active by default, and duplicate events by user, item, and timestamp are removed. The planned evaluation follows leave-one-out splitting: the last item in each eligible user sequence is used for testing, the second-to-last item for validation, and earlier items for training.

Results

[TBD: This subsection will report the final Recall@K, nDCG@K, HitRate@K, diversity, and novelty results after model training and evaluation are completed.]

Contributions

The key contributions of this work include:

1. A data-driven problem formulation for next learning resource recommendation on MARS.
2. An EDA summary identifying sparsity, sequence-length, popularity-skew, and metadata-coverage constraints.

3. A planned experimental pipeline covering baselines, gSASRec, ablations, and a lightweight demo-oriented output.

Keywords: Sequential recommendation, gSASRec, e-learning, MARS dataset, personalized learning

Mục lục

1	Introduction	6
1.1	Problem Statement	6
1.2	Objectives and Scope	6
1.3	Research Motivation	7
1.4	Dataset and Model Direction	7
1.5	Thesis Contribution	7
1.6	Expected Output and Team Responsibilities	8
1.7	Thesis Organization	8
2	Background	9
2.1	Recommendation Systems Fundamentals	9
2.1.1	Collaborative Filtering	9
2.1.2	Content-Based Filtering	9
2.2	Sequential Recommendation Models	9
2.2.1	Recurrent Neural Networks (RNNs)	9
2.2.2	Self-Attention Mechanisms	9
2.3	Graph-Based Sequential Recommendation	9
2.3.1	The gSASRec Model	9
2.4	The MARS E-Learning Dataset	10
3	Related Work	11
3.1	Sequential Recommendation Systems	11
3.1.1	RNN-Based Approaches	11
3.1.2	Attention-Based Methods	11
3.1.3	Graph-Enhanced Approaches	11
3.2	Sequential Recommendation in E-Learning	11
3.3	Dataset Studies	12
3.4	Performance Metrics	12
3.5	Gap Analysis	12
4	Methodology	13
4.1	Methodology Overview	13
4.2	Data Preprocessing	13
4.2.1	Data Loading	13
4.2.2	Interaction Construction	13
4.2.3	Sequence Handling	13
4.2.4	Filtering Strategy	14
4.2.5	Evaluation Split	14
4.3	Model Direction	14
4.3.1	Baselines	14
4.3.2	SASRec Backbone	14
4.3.3	gSASRec Direction	14
4.3.4	Metadata and Confidence Extensions	15

4.4	Model Training	15
4.4.1	Hyperparameters	15
4.5	Evaluation Methodology	15
5	Experiments	16
5.1	Experimental Setup	16
5.1.1	Dataset Statistics	16
5.1.2	Implementation Details	17
5.1.3	Hyperparameter Tuning	18
5.2	Results	18
5.2.1	Main Performance Results	18
5.2.2	Ablation Study	18
5.3	Analysis	18
5.3.1	Performance Insights	18
5.3.2	Learning Dynamics	19
5.3.3	Error Analysis	19
5.4	Statistical Significance	19
6	Conclusion	20
6.1	Summary of Contributions	20
6.2	Key Findings	20
6.3	Limitations	20
6.4	Future Work	21
6.5	Broader Impact	21
6.6	Final Remarks	21

1. Introduction

The rapid growth of online learning platforms has created unprecedented opportunities for students and employees to access educational content at scale. However, a large catalogue of videos, courses, and instructional documents can also create *choice overload*: learners may know the general topic they want to study, but still struggle to choose the next useful resource. For platform operators and educators, this creates a practical need for recommendation systems that can adapt to each learner’s interaction history and surface relevant next resources.

This thesis focuses on the MARS (Mandarine Academy Recommender System) dataset, a real-world corporate e-learning dataset from Mandarine Academy. The project frames the task as sequential top- K recommendation: given a learner’s historical sequence of resource interactions, recommend the most relevant learning resources that should be considered next.

1.1. Problem Statement

The problem addressed in this thesis is:

Given a learner u and the chronological interaction sequence $S_u = (i_1, i_2, \dots, i_t)$, predict a ranked list of K candidate learning resources $(\hat{i}_{t+1}^1, \dots, \hat{i}_{t+1}^K)$ that are likely to match the learner’s next interaction.

This is challenging in the MARS setting for several reasons:

- The active interaction matrix is extremely sparse, with only 0.8547% observed user–item pairs and 99.1453% sparsity.
- Learner histories are short for many users; only 1,676 out of 3,007 users have at least three interactions for leave-one-out evaluation.
- Item popularity is highly skewed, meaning a small number of resources can dominate naive recommendation results.
- Educational resources have useful but incomplete metadata; difficulty is available for only 40.70% of items, while descriptions are available for 86.46%.
- Static recommenders lose the temporal structure of learning behavior, while neural sequential recommenders can be sensitive to sparse data and negative sampling choices.

1.2. Objectives and Scope

The main objectives of the thesis are:

1. Build a reproducible data pipeline for loading, validating, preprocessing, and analyzing the MARS dataset.
2. Define a sequential next-resource recommendation task using chronological user histories and leave-one-out evaluation.
3. Implement and evaluate baseline methods including popularity-based recommendation, item-based collaborative filtering, BPR-MF, and SASRec.

4. Develop a gSASRec-oriented model direction that uses Generalised BCE loss, multiple negative samples, and available item metadata where appropriate.
5. Measure recommendation quality using Recall@K, nDCG@K, HitRate@K, and practical ranking indicators such as diversity and novelty.

The scope is limited to offline recommendation experiments and a prototype-scale demonstration. The thesis does not claim to measure real learning gain, user satisfaction, or production-level serving performance because these require online deployment and learner feedback. Cold-start handling is considered through metadata-aware item representations, but full personalization for completely new users is outside the core scope.

1.3. Research Motivation

The MARS dataset represents a realistic recommendation scenario because it combines implicit interaction logs, explicit ratings or watch signals, item metadata, and user profile information. The current EDA highlights several properties that make the dataset academically and practically interesting:

- 3,007 users, 958 items, and 24,622 active interactions after preprocessing.
- An average of 8.19 interactions per user, with a wide range from 1 to 395.
- An average of 25.70 users per item, with a maximum of 886 users for the most popular resource.
- Complete coverage for software, theme, duration, and type metadata, enabling content-aware extensions.

These characteristics motivate a model direction that combines sequential modeling with careful loss design and negative sampling, instead of relying only on popularity or static user-item affinity.

1.4. Dataset and Model Direction

The dataset consists of four main sources: implicit interactions, explicit interactions, item metadata, and user information. The preprocessing pipeline merges implicit and explicit interactions, removes duplicate events by user, item, and timestamp, and keeps active interactions for the main EDA. Explicit events with watch percentage greater than or equal to 50 are treated as active, while implicit events are treated as active by default.

The intended modeling direction is to use gSASRec as the primary sequential recommender. The baseline SASRec architecture models the learner’s ordered history with self-attention. The gSASRec direction keeps the Transformer-style sequential architecture but focuses on Generalised BCE loss and multiple negative sampling to reduce overconfidence under sparse, large-catalogue conditions. Additional planned components include confidence weighting from watch completion, metadata-based item embedding initialization, and reranking for diversity and novelty.

1.5. Thesis Contribution

The expected contributions of this thesis are:

- A clear problem formulation for next learning resource prediction on the MARS dataset.
- A reproducible EDA and preprocessing workflow for sparse e-learning recommendation data.

- A systematic comparison between non-neural, matrix-factorization, SASRec, and gSASRec-oriented approaches.
- An offline evaluation report and prototype output that can support future deployment work.

1.6. Expected Output and Team Responsibilities

The expected project outputs are:

- Source code for preprocessing, EDA, model training, evaluation, and inference.
- EDA artifacts including summary tables, plots, and dataset reports.
- Trained model checkpoints and configuration files for reproducible experiments.
- Offline evaluation tables for Recall@K, nDCG@K, HitRate@K, diversity, and novelty.
- A prototype recommendation demo, planned with Streamlit and a Python/FastAPI backend at demo scale.
- Final thesis report, defense presentation, and supporting documentation.

The planned team responsibilities are:

Bảng 1.1: Planned team responsibilities	
Member	Primary responsibilities
Nguyễn Tấn Thắng	Team lead; project coordination; evaluation protocol; experiment tracking; final integration and presentation preparation.
Nguyễn Hoàng Việt	Data pipeline; EDA; preprocessing; dataset documentation; thesis writing for problem statement, data analysis, and methodology.
Lê Quốc Chính	Baseline implementation; gSASRec training support; demo application; result visualization and deployment documentation.

1.7. Thesis Organization

The remainder of this thesis is organized as follows:

- **Chapter 2** provides background on recommendation systems and sequential models
- **Chapter 3** reviews related work in sequential recommendation and e-learning
- **Chapter 4** details our methodology and model architecture
- **Chapter 5** presents experimental setup and results
- **Chapter 6** concludes with key findings and future research directions

2. Background

2.1. Recommendation Systems Fundamentals

Recommendation systems are computational methods that predict user preferences or ratings for items they have not yet interacted with. They are widely used in e-commerce, streaming platforms, social media, and increasingly in educational settings.

2.1.1 Collaborative Filtering

Collaborative filtering is based on the assumption that users with similar preferences in the past will have similar preferences in the future. It includes:

- **User-based:** Find similar users and recommend items they liked
- **Item-based:** Find similar items and recommend based on user history

2.1.2 Content-Based Filtering

Content-based approaches recommend items similar to those the user has liked before, using item features and user profiles.

2.2. Sequential Recommendation Models

Sequential models consider the temporal ordering and dependencies in user interactions, which is crucial for domains like e-learning.

2.2.1 Recurrent Neural Networks (RNNs)

RNNs, including variants like LSTM and GRU, naturally capture sequential dependencies through their recurrent connections.

2.2.2 Self-Attention Mechanisms

Self-attention allows models to weigh the importance of different historical items when making recommendations, providing better interpretability than traditional RNNs.

2.3. Graph-Based Sequential Recommendation

Graph structures can represent relationships between users, items, and sequences. Graph neural networks (GNNs) combined with sequential models like gSASRec provide a powerful framework.

2.3.1 The gSASRec Model

The gSASRec (Graph Self-Attentive Sequential Recommendation) model combines:

1. Graph neural network layers to capture item relationships
2. Self-attention mechanisms to model temporal dependencies
3. Sequence modeling to leverage historical interactions

2.4. The MARS E-Learning Dataset

The MARS (Massive Open Online Courses) e-learning dataset contains:

- Real student interaction logs from online courses
- Learning resource metadata and relationships
- Temporal information about when students access resources
- Multiple resource types (videos, quizzes, documents, etc.)

This dataset presents a realistic testbed for sequential recommendation research with domain-specific characteristics.

3. Related Work

3.1. Sequential Recommendation Systems

Sequential recommendation has become increasingly important as researchers recognize that user behavior is inherently temporal and sequential in nature.

3.1.1 RNN-Based Approaches

Early works used RNNs to capture sequential patterns:

- Hidasi et al. introduced GRU4REC, using gated recurrent units for session-based recommendations
- Extensions like improved GRU4REC added data augmentation and techniques to address data sparsity

3.1.2 Attention-Based Methods

The introduction of self-attention in sequential recommendation improved performance and interpretability:

- SASRec (Self-Attentive Sequential Recommendation) proposed by Kang et al. uses multi-head self-attention
- Subsequent works extended this with graph structures and improved architectures

3.1.3 Graph-Enhanced Approaches

Recent methods combine graphs with sequential models:

- gSASRec augments SASRec with item relationships through graph embeddings
- Other works explore heterogeneous graphs for multi-type interactions

3.2. Sequential Recommendation in E-Learning

E-learning systems have unique characteristics that affect recommendation design:

- Learning outcomes depend on prerequisites and learning sequences
- Pedagogical constraints must be considered (prerequisites, difficulty progression)
- Students have diverse learning goals and backgrounds
- Course structures impose constraints on valid recommendation sequences

Few works specifically target sequential recommendation for e-learning platforms, making this an important research direction.

3.3. Dataset Studies

Research on the MARS dataset and similar e-learning datasets:

- MARS provides opportunities to study real student behavior patterns
- Prior analysis has identified key characteristics like resource popularity skew
- Temporal patterns in student access reveal learning rhythm variations

3.4. Performance Metrics

Common metrics for evaluating sequential recommendation systems include:

1. **Recall@K:** Proportion of relevant items in top-K recommendations
2. **NDCG@K:** Normalized discounted cumulative gain considering ranking quality
3. **MRR:** Mean reciprocal rank of the first relevant item
4. **Hit Rate:** Frequency with which correct item appears in top-K

3.5. Gap Analysis

Despite the growing attention to sequential recommendation and e-learning systems, there remains a gap in:

- Specialized models for educational domain constraints
- Comprehensive evaluation on real e-learning datasets like MARS
- Practical deployment considerations for production systems

This thesis addresses these gaps by proposing and evaluating a gSASRec-based system specifically for the MARS e-learning context.

4. Methodology

4.1. Methodology Overview

The methodology is designed around an offline sequential recommendation pipeline. The system transforms raw MARS interaction logs into chronological user sequences, trains baseline and neural recommendation models, evaluates them with leave-one-out ranking metrics, and exposes the final model through a prototype recommendation interface.

The planned pipeline consists of five components:

1. **Data preprocessing:** Load, validate, merge, clean, and filter interaction data.
2. **Exploratory data analysis:** Quantify sparsity, sequence length, popularity skew, temporal patterns, and metadata coverage.
3. **Model training:** Train popularity, CF, BPR-MF, SASRec, and gSASRec-oriented models.
4. **Evaluation:** Apply leave-one-out evaluation with top- K metrics and ablation studies.
5. **Prototype output:** Serve recommendations through a lightweight demo interface.

4.2. Data Preprocessing

4.2.1 Data Loading

The MARS dataset is loaded from four source files:

- Implicit interaction logs.
- Explicit ratings and watch-completion signals.
- Item metadata, including theme, software, duration, type, difficulty, and text description.
- User profile information.

The EDA code validates the required columns for implicit interactions, explicit interactions, and item metadata before downstream processing.

4.2.2 Interaction Construction

Implicit and explicit events are merged into a unified interaction table. Duplicate events with the same user, item, and timestamp are dropped. Explicit interactions with watch percentage greater than or equal to 50 are treated as active positive feedback; implicit interactions are treated as active by default. The resulting active interaction table is the main input for sequence construction.

4.2.3 Sequence Handling

For each learner, interactions are sorted chronologically to construct a resource sequence. Users with fewer than three interactions are excluded from leave-one-out train/validation/test splitting because they cannot provide at least one training item, one validation item, and one test item. According to the current EDA, 1,676 users satisfy this criterion.

4.2.4 Filtering Strategy

The EDA pipeline reports both the unfiltered active dataset and k -core filtering experiments:

- Active dataset: 3,007 users, 958 items, and 24,622 interactions.
- 3-core dataset: 1,673 users, 802 items, and 22,557 interactions.
- 5-core dataset: 1,116 users, 658 items, and 20,150 interactions.

The final training configuration will choose the filtering level based on validation performance and the trade-off between interaction density and catalogue/user coverage.

4.2.5 Evaluation Split

The planned evaluation protocol is leave-one-out. For each eligible user, the last item is held out for testing, the second-to-last item is held out for validation, and the remaining earlier interactions are used for training. This protocol preserves chronological order and avoids using future interactions to predict past behavior.

4.3. Model Direction

4.3.1 Baselines

The project evaluates the proposed direction against several baselines:

1. **Popularity:** Recommends globally popular learning resources.
2. **Item-based collaborative filtering:** Recommends resources similar to those in the learner’s history.
3. **BPR-MF:** Learns static user–item affinity from implicit feedback.
4. **SASRec:** Uses self-attention to model ordered user histories.

4.3.2 SASRec Backbone

SASRec represents each item in the sequence with item and positional embeddings, then applies stacked self-attention blocks to model dependencies between historical interactions. The attention operation is:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

The final hidden representation is compared against candidate item embeddings to produce a ranked recommendation list.

4.3.3 gSASRec Direction

The primary model direction is gSASRec, which addresses overconfidence in sequential recommendation by replacing standard BCE training with Generalised BCE and by using multiple negative samples per positive interaction. This is especially relevant to MARS because the dataset is highly sparse and has a moderately large item catalogue relative to the number of observed interactions.

The planned loss direction can be summarized as:

$$\mathcal{L}_{gBCE} = -w^+ \log \sigma(s^+/\tau) - \sum_{j=1}^m w_j^- \log (1 - \sigma(s_j^-/\tau)),$$

where s^+ is the positive item score, s_j^- is the score of the j -th negative item, m is the number of negative samples, τ is the temperature parameter, and w^+ and w_j^- are optional confidence weights.

4.3.4 Metadata and Confidence Extensions

Item metadata will be used where it is reliable enough to support cold-start mitigation and candidate analysis. Since software, theme, duration, and type have 100% coverage, these features are suitable for metadata-aware item representations. Difficulty has only 40.70% coverage and will be handled carefully to avoid introducing missing-value bias. Watch-completion percentage can be used as a soft confidence signal for explicit positive interactions.

4.4. Model Training

4.4.1 Hyperparameters

The planned hyperparameter search includes:

- Embedding dimensions: 32, 64, and 128.
- Number of attention heads: 1, 2, 4, and 8.
- Number of attention layers: 1, 2, and 3.
- Learning rates: 0.001, 0.0005, and 0.0001.
- Negative sample counts and temperature values for the gSASRec loss.
- Sequence length cutoff and k -core threshold.

[TBD: The final selected hyperparameters will be inserted after validation experiments.]

4.5. Evaluation Methodology

The main evaluation metrics are:

- **Recall@K:** Whether the held-out item is retrieved in the top- K list.
- **nDCG@K:** Ranking quality with higher weight for correct items near the top.
- **HitRate@K:** Whether at least one ground-truth item appears in the top- K list.
- **Diversity:** Variety among recommended resources.
- **Novelty:** Ability to recommend useful resources beyond the most popular items.

The ablation study will isolate the effect of Generalised BCE, number of negative samples, temperature, metadata initialization, confidence weighting, and reranking.

5. Experiments

5.1. Experimental Setup

5.1.1 Dataset Statistics

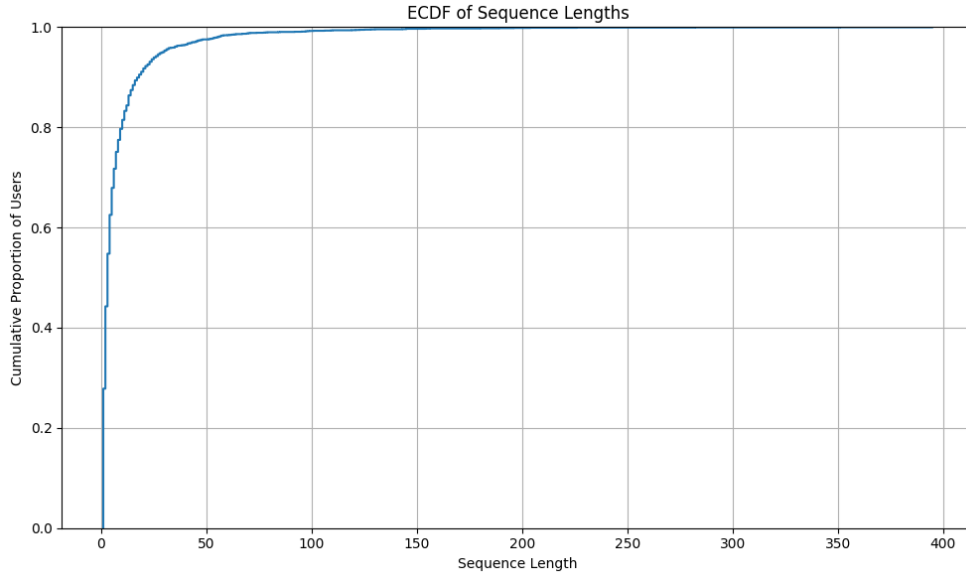
The current EDA was generated by the pipeline in `src/mars_rec/eda`. It loads the MARS interaction and metadata files, validates the required schema, merges implicit and explicit interactions, filters active interactions, computes dataset statistics, and produces plots in `outputs/eda/figures`.

Bảng 5.1: MARS active interaction statistics from EDA

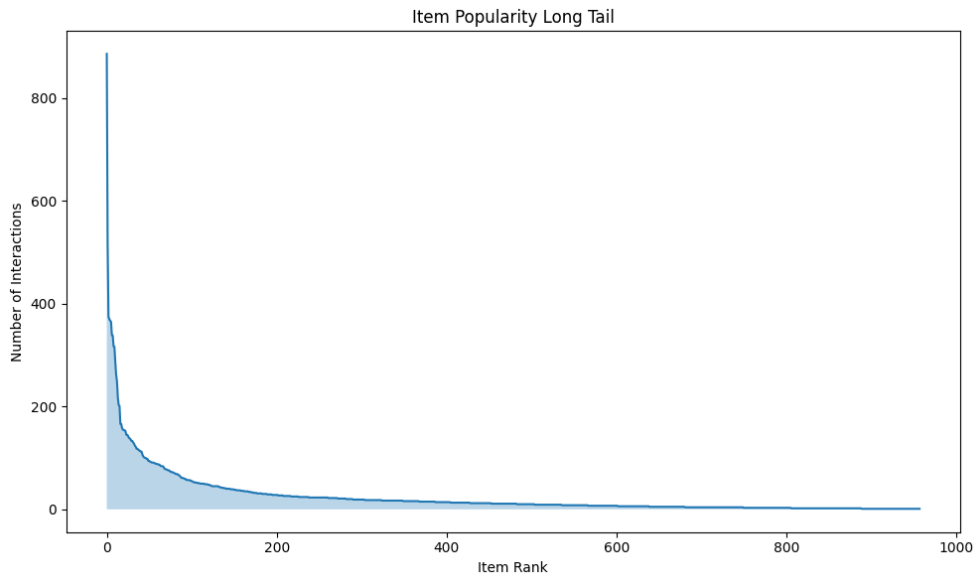
Statistic	Value
Users	3,007
Items	958
Active interactions	24,622
Sparsity	99.1453%
Density	0.8547%
Mean interactions per user	8.19
Min/max interactions per user	1 / 395
Mean users per item	25.70
Min/max users per item	1 / 886
Users with sequence length ≥ 3	1,676

Bảng 5.2: Item metadata coverage

Metadata field	Coverage
Difficulty	40.70%
Software	100.00%
Theme	100.00%
Duration	100.00%
Type	100.00%
Description	86.46%



Hình 5.1: Sequence length distribution in the active interaction dataset



Hình 5.2: Long-tail item popularity pattern

5.1.2 Implementation Details

The planned implementation stack is:

- **Language:** Python.
- **Model framework:** PyTorch.
- **Data processing:** pandas and parquet intermediate files.
- **Demo:** Streamlit with a Python/FastAPI backend at prototype scale.
- **Database:** SQLite for lightweight demo storage.

5.1.3 Hyperparameter Tuning

We conducted grid search over:

- Embedding dimensions: {32, 64, 128}
- Number of attention heads: {1, 2, 4, 8}
- Number of layers: {1, 2, 3}
- Learning rates: {0.001, 0.0005, 0.0001}

Optimal hyperparameters were selected based on validation set performance. [TBD: Replace this sentence with the final selected hyperparameters and validation criterion after the training runs are completed.]

5.2. Results

5.2.1 Main Performance Results

Bảng 5.3: Sequential Recommendation Performance Comparison

Method	Recall@5	NDCG@5	MRR@5	Hit Rate@5
Popularity	[TBD]	[TBD]	[TBD]	[TBD]
Item-based CF	[TBD]	[TBD]	[TBD]	[TBD]
BPR-MF	[TBD]	[TBD]	[TBD]	[TBD]
SASRec	[TBD]	[TBD]	[TBD]	[TBD]
gSASRec Direction	[TBD]	[TBD]	[TBD]	[TBD]

[TBD: Insert final quantitative results after model training, validation, and test evaluation.]

5.2.2 Ablation Study

To understand the contribution of different components, the planned ablation study will compare the following variants:

Bảng 5.4: Ablation Study Results

Model Variant	Recall@5
gSASRec direction (full setting)	[TBD]
Standard BCE instead of Generalised BCE	[TBD]
Single negative sample instead of multiple negatives	[TBD]
Without metadata initialization	[TBD]
Without confidence weighting	[TBD]
Without diversity/novelty reranking	[TBD]

5.3. Analysis

5.3.1 Performance Insights

[TBD: To be filled with detailed analysis of model performance, including which model performs best under Recall@K, nDCG@K, and HitRate@K.]

5.3.2 Learning Dynamics

[TBD: To be filled with training curves, convergence behavior, validation performance, and early-stopping observations.]

5.3.3 Error Analysis

The expected error analysis will examine:

- Cold-start or low-history users.
- Rare resources in the long tail.
- Cases where popularity dominates personalized signals.
- Sequences with unusual or non-linear learning patterns.

5.4. Statistical Significance

[TBD: Add significance testing or confidence intervals if repeated runs are performed. Otherwise, state the residual evaluation risk clearly.]

6. Conclusion

6.1. Summary of Contributions

This thesis studies next learning resource prediction on the MARS e-learning dataset. At the current writing stage, the completed contribution is the problem formulation, dataset analysis, planned methodology, and experimental design. The final model results will be added after training and evaluation are completed.

1. **Problem formulation:** The project defines MARS recommendation as a sequential top- K next-item prediction task.
2. **Dataset analysis:** The EDA identifies high sparsity, short user histories, long-tail item popularity, and partial metadata coverage.
3. **Experimental design:** The planned evaluation compares popularity, item-based collaborative filtering, BPR-MF, SASRec, and a gSASRec-oriented direction.

6.2. Key Findings

The current data analysis shows that:

- The active MARS dataset contains 3,007 users, 958 items, and 24,622 interactions.
- The interaction matrix is highly sparse, with 99.1453% sparsity.
- Only 1,676 users have at least three interactions and are eligible for leave-one-out evaluation.
- Metadata is useful but uneven: software, theme, duration, and type are complete, while difficulty is available for 40.70% of items.

[TBD: Add final findings about model performance after experiments are completed.]

6.3. Limitations

While promising, this work has several limitations:

- Evaluation is limited to a single dataset; generalization to other e-learning platforms remains uncertain.
- Many users have short histories, which limits sequential signal and makes personalization difficult.
- Difficulty metadata is incomplete, so metadata-aware modeling must handle missing values carefully.
- Offline metrics do not directly prove improved learning outcomes or learner satisfaction.
- Prototype-scale deployment does not fully address large-scale production serving constraints.

6.4. Future Work

Potential extensions of this research include:

1. **Multi-Modal Resources:** Extending the model to handle diverse resource types (videos, documents, quizzes, assignments)
2. **Cross-Platform Generalization:** Evaluating on multiple e-learning datasets to assess generalization
3. **Real-Time Recommendations:** Developing online learning algorithms for practical deployment
4. **Explainability:** Adding interpretability mechanisms to explain why specific resources are recommended
5. **Pedagogical Constraints:** Incorporating learning objectives and prerequisite information more explicitly
6. **User Diversity:** Personalization based on learning styles, backgrounds, and goals

6.5. Broader Impact

This work contributes to the growing field of intelligent e-learning systems. By improving resource recommendations, we aim to:

- Enhance student learning outcomes and engagement
- Reduce cognitive load through curated learning sequences
- Support educators in creating more effective learning experiences
- Enable broader access to quality educational resources through better discovery

6.6. Final Remarks

Sequential recommendation in e-learning is an important and practical research direction. The MARS dataset provides a challenging testbed because it combines real learner behavior, sparse interactions, short sequences, and useful but incomplete metadata. [TBD: Replace this placeholder with the final conclusion after all experiments are complete.]

References

[TBD: Add complete BibTeX entries and citations for SASRec, gSASRec, BPR-MF, GRU4Rec, TiSAS-Rec, MARS dataset, and evaluation metrics before the final submission.]